

RETAIL DISCOUNT PROFITABILITY & REVENUE LEAKAGE AUDIT

Executive Findings & Strategic Assessment

Dataset: Contoso Retail (Synthetic) • Scope: ~224K Order Lines • ~52K Customers
Coverage: Multi-Year Transaction History • Currency: Multi (normalization not applied)

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PROJECT	Retail Discount Profitability & Revenue Leakage Audit
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STATUS	Final • Internal Review

POSTGRESQL	SQL audit layer • Transaction-level profitability analysis
PYTHON	Pandas, Matplotlib • Validation and customer behavioral modeling
POWER BI	Executive dashboards • 3-page operational reporting suite

This document is prepared for internal strategy review. Findings are based on the Contoso retail synthetic dataset and should not be applied to live operational decisions without validation against actual business data.

1. Executive Summary

Discounting is not functioning as a tactical pricing lever. It is the company's structural pricing floor, applied at scale across all categories, stores, and customer segments, with no sign of tactical targeting or promotional logic.

TOTAL REVENUE	BLENDED MARGIN	DISC. PENETRATION	DISCOUNTED MARGIN	PROFIT GAP / LINE
\$218.8M	55.9%	61.1%	54.0%	~\$97
~224K order lines	Full price: 58.1%	Of all transactions	vs 58.1% full price	vs full-price lines

Key Findings

- **Discount Prevalence.** Approximately 61.1% of all order lines are transacted below list price. Discounting is the majority pricing behavior, not a promotional exception.
- **Margin Compression.** Margin falls consistently from 53.9% in the low-discount tier to 48.9% in the high-discount tier. The relationship is monotonic with no tier showing recovery.
- **Profit Efficiency.** Discounted transactions generate approximately \$97 less profit per order line than full-price transactions. At 61% penetration, this represents a persistent margin drag across the entire transaction base.
- **Category Concentration.** Computers and Cell Phones contribute nearly half of all high-discount volume. Their outsized aggregate impact is explained by transaction scale, not by more aggressive discounting. Compression rates are similar across all categories.
- **Customer Dependency.** Approximately 24.2% of customers transact at or above 90% discount usage. High-discount customers produce the lowest margins and the lowest purchase frequency of any segment.
- **Structural Stability.** Discount rates and margin levels show no deterioration year-over-year. The pattern reflects a stable, structurally embedded pricing model rather than a deteriorating promotional trend.

Business Interpretation

The analysis indicates that discounting has transitioned from a tactical sales mechanism into the company's de facto pricing floor. Because the pattern is consistent across categories, stores, and years, the current blended margin of approximately 55.9% represents the actual operating baseline under this model, not a temporary suppression of a healthier one. Customers in the high-dependency segment transact at a discount roughly 96% of the time, indicating that discounted pricing has become their default expectation.

Operational Risk: Because margin compression is structurally uniform and shows no acceleration year-over-year, management may underestimate its long-term cumulative impact. As transaction volume grows, total margin loss compounds in direct proportion to scale. Corrective intervention must occur at the policy level to produce measurable results.

2. Analytical Methodology

The audit was conducted across three analytical layers. SQL provided the structural profitability baseline. Python extended the analysis to behavioral patterns at the customer level and independently validated SQL findings. Power BI translated the findings into an executive dashboard format for operational review.

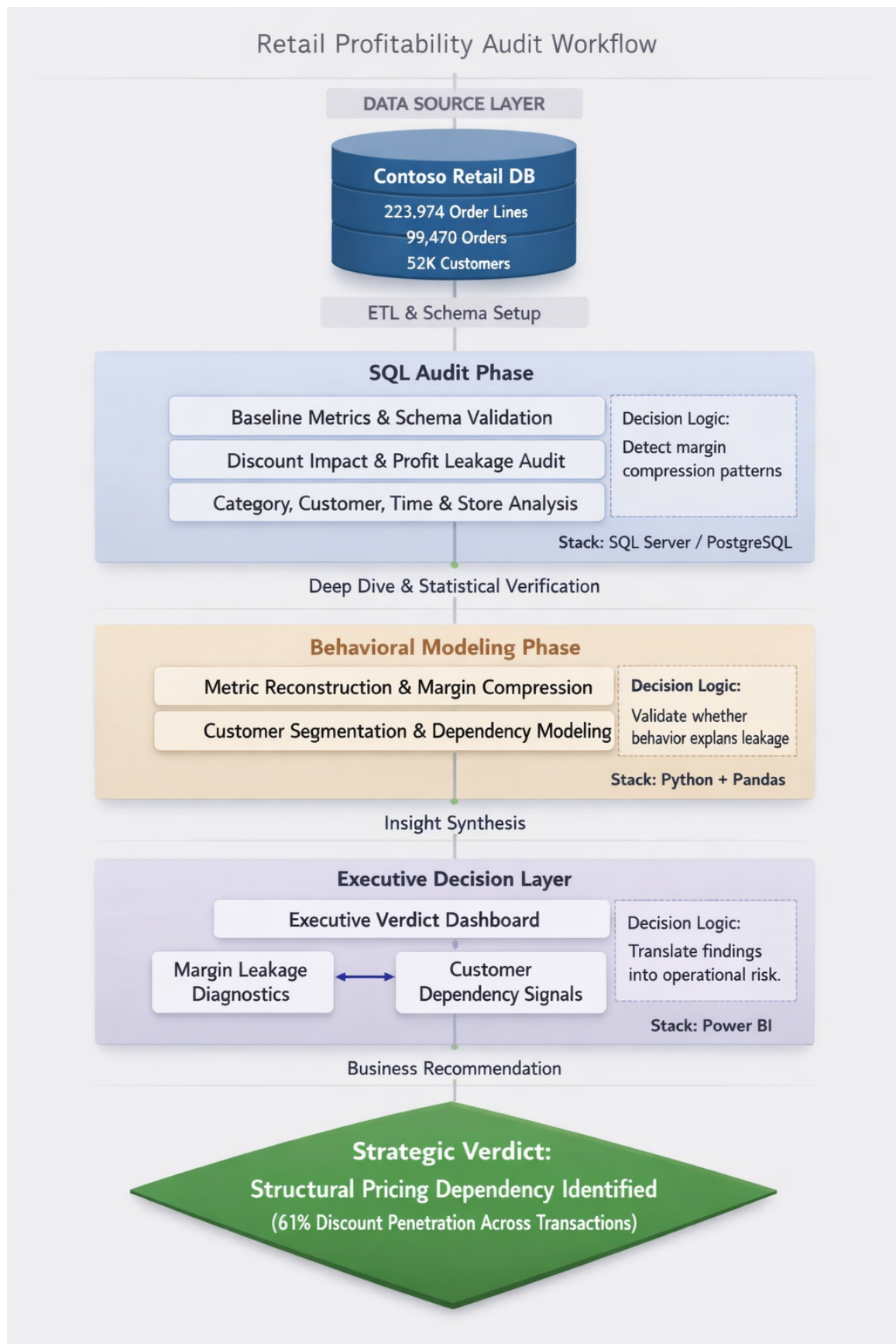


Figure 1. Analytical workflow architecture.

Analytical Layers

LAYER	TOOLS	PURPOSE
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SQL Audit	PostgreSQL v18.3	Transaction reconstruction, discount tier analysis, category and store validation, year-over-year trends
Python Validation	Pandas, Matplotlib	Metric reconciliation, customer behavioral segmentation, distribution analysis, category impact scoring
Power BI Reporting	Power BI Desktop	3-page executive dashboard, KPI monitoring, decision-oriented presentation layer

Business Questions Answered

- Is discounting reducing overall profitability, and by how much?
- Does margin compress as discount depth increases, and is the relationship consistent?
- Which product categories drive the largest aggregate margin leakage?
- Are high-discount customers commercially more or less valuable than low-discount customers?
- Is pricing behavior centrally standardized, or do stores exercise independent discretion?
- Has discount behavior worsened over time, or is it structurally stable?

Dataset and Scope

The analysis used the Contoso retail synthetic dataset. The primary analytical grain was the order line level. Six tables were used: orderrows (transaction grain), orders (order header), customer, product, store, and date dimension. The currency exchange table was present in the schema but not applied, as the analysis focused on relative margin metrics.

METRIC	VALUE
Order Lines	223,974
Orders	93,470
Customers	52,189
Products	2,517
Stores	74
Time Coverage	Multi-year (2016 onward)

3. Transaction-Level Profitability Audit

The transaction-level audit established the core economic relationship between discounting and profitability. All metrics are computed at the order line level using quantity-adjusted values, consistent with how discounts are applied in practice.

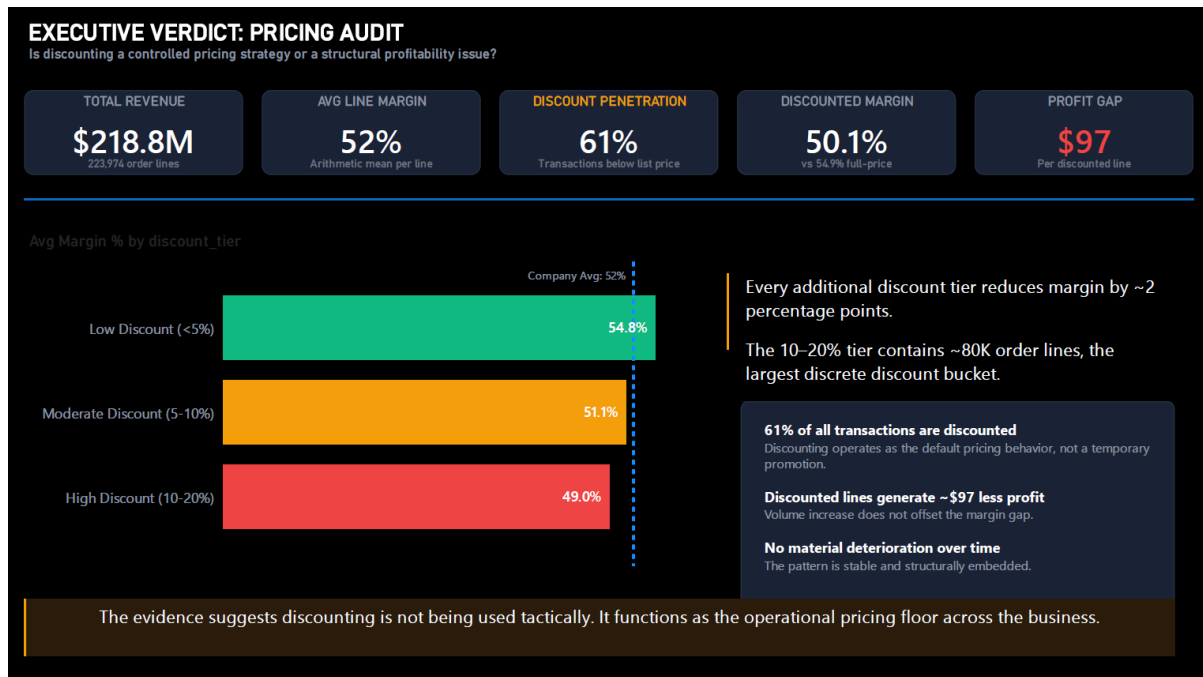


Figure 2. Power BI Page 1 Executive Dashboard.

Discount Penetration

Approximately 61.1% of all order lines are transacted below the listed unit price. At this rate, discounting is the dominant pricing behavior across the transaction base. A selective promotional strategy would be expected to produce a materially lower penetration rate.

Discounted vs. Full-Price Performance

METRIC	DISCOUNTED	FULL PRICE	DIFFERENCE
Avg Margin %	54.0%	58.1%	-4.1 pts
Profit per Line	~\$508	~\$605	-\$97
Order Line Volume	~136,800 (61%)	~87,200 (39%)	—

Margin Compression by Discount Tier

Discounted transactions were segmented by discount depth. Margin declines at every step. No tier shows recovery. The high-discount tier (10-20%) accounts for the majority of discounted volume. No transactions in the dataset exceed 20% discount depth, confirming that compression is driven by mid-range discounts applied at high volume, not by extreme reductions.

DISCOUNT TIER	ORDER LINES	AVG DISCOUNT	AVG MARGIN
Low (<5%)	14,701	2.6%	53.9%
Moderate (5-10%)	42,288	7.9%	51.1%
High (10-20%)	79,924	11.9%	48.9%
Deep (>20%)	0	N/A	N/A

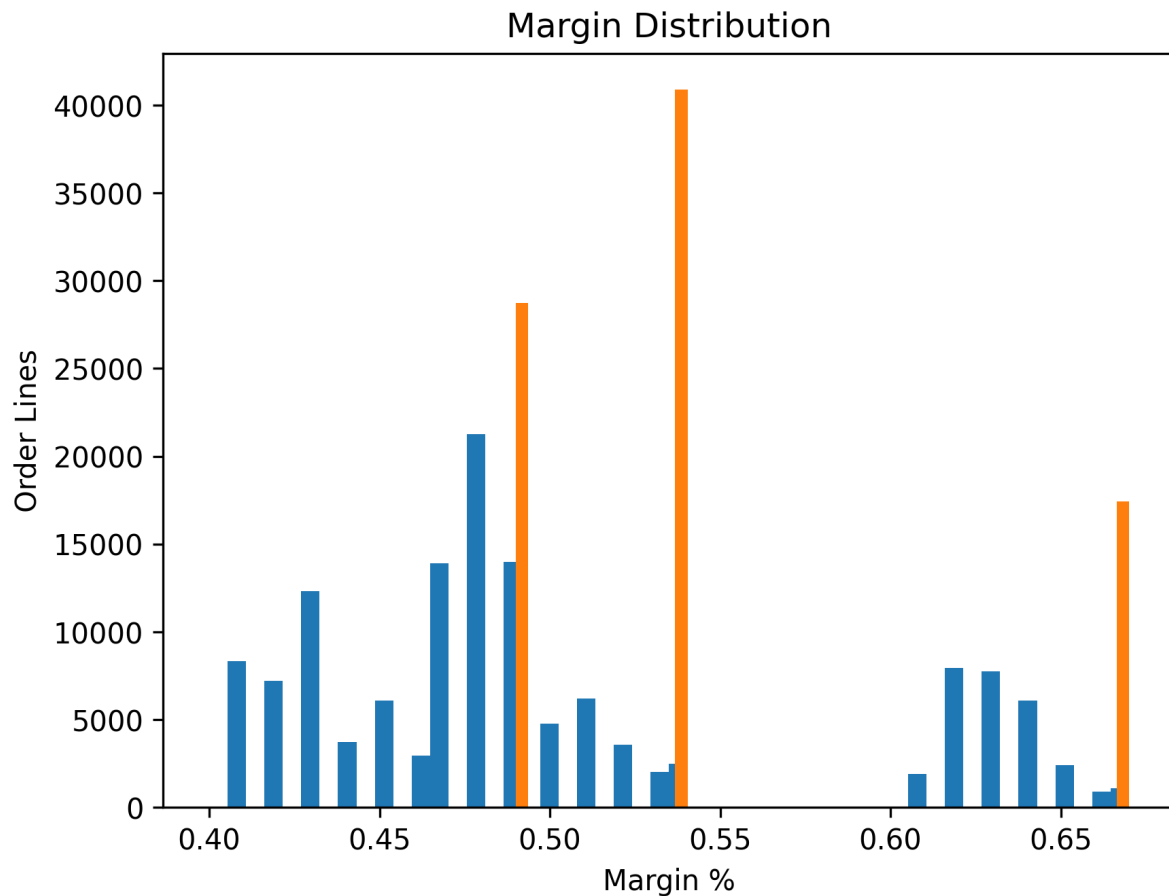


Figure 3. Margin distribution comparison.

Audit Finding: The issue is not isolated discount misuse. Discounted pricing has become the operational norm, applied consistently across the majority of transactions regardless of category, customer, or time period.

4. Category & Revenue Leakage Analysis

The category-level analysis identified where margin leakage concentrates in aggregate terms. Per-unit margin compression is broadly uniform across all categories (approximately 4.9 to 5.3 percentage points). Total leakage per category is therefore determined almost entirely by transaction volume in the high-discount tier.



Figure 4. Power BI Page 2 Leakage Diagnostics.

Category Leakage Summary

CATEGORY	HIGH-DISC LINES	MARGIN DROP	IMPACT SCORE
Computers	18,074	4.9 pts	892.8
Cell Phones	17,366	5.1 pts	885.7
Music / Movies / Audio	13,794	4.9 pts	677.3
Games and Toys	8,164	5.3 pts	431.9
Home Appliances	6,785	5.0 pts	342.0
Audio	5,527	5.1 pts	284.1
Cameras & Camcorders	4,979	5.3 pts	265.9
TV and Video	5,235	4.9 pts	256.0

Impact Score = margin drop (pct pts) x high-discount order lines. This is a relative index, not a direct monetary loss figure.

Interpretation

Computers and Cell Phones together account for approximately 46% of all high-discount order lines. Their combined impact score is approximately 1,778, nearly three times the next closest category. This concentration is not the result of deeper or more aggressive discounting. Margin compression rates are nearly identical across all categories, ranging from 4.9 to 5.3 percentage points. The gap is driven entirely by transaction scale.

The business is not losing margin because certain categories are mispriced. It is losing margin because standardized discounting is applied at large transactional scale, and the categories with the highest volume absorb the largest share of that compression.

5. Customer Behavioral Analysis

The customer-level analysis examined whether discount dependency is explained by individual customer behavior or by a structurally applied pricing mechanism. Customers were segmented by discount dependency rate: the proportion of their transactions that occurred below list price.

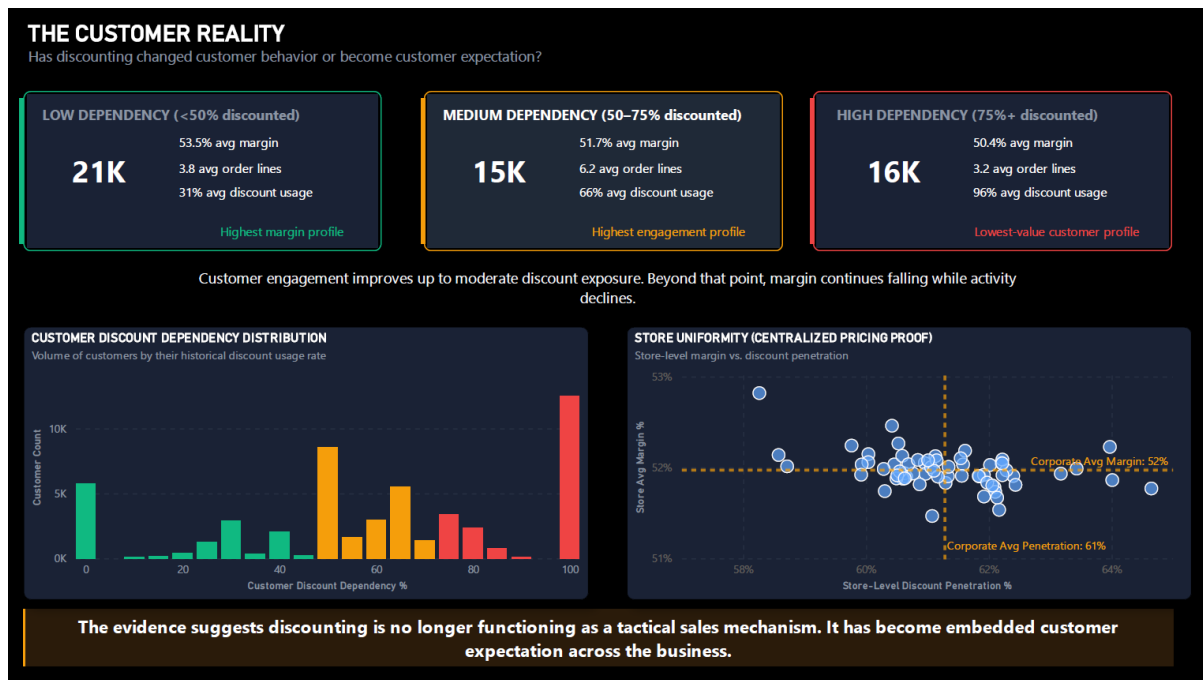


Figure 5. Power BI Page 3 Customer Reality.

Customer Discount Dependency Segments

SEGMENT	CUSTOMERS	AVG DISC RATE	AVG MARGIN	AVG ORDER LINES
Low (<50% disc.)	21,336	30.9%	53.5%	3.77
Medium (50-75% disc.)	14,758	66.4%	51.7%	6.23
High (>75% disc.)	16,095	96.1%	50.4%	3.20

Behavioral Findings

- **Margin decline is monotonic across segments.** Average margin falls from 53.5% (Low) to 51.7% (Medium) to 50.4% (High). The 3.1 point gap between extremes confirms that discount dependency correlates directly with lower profitability at the customer level.
- **Engagement follows an inverted-U pattern.** Medium-segment customers generate 6.23 average order lines, nearly double the High segment at 3.20. Discounting increases purchase activity up to a threshold, beyond which additional dependency reduces engagement rather than increasing it.
- **High-discount customers are not commercially efficient.** They produce the lowest margins and the lowest purchase frequency simultaneously. Margin erosion is not offset by volume compensation.

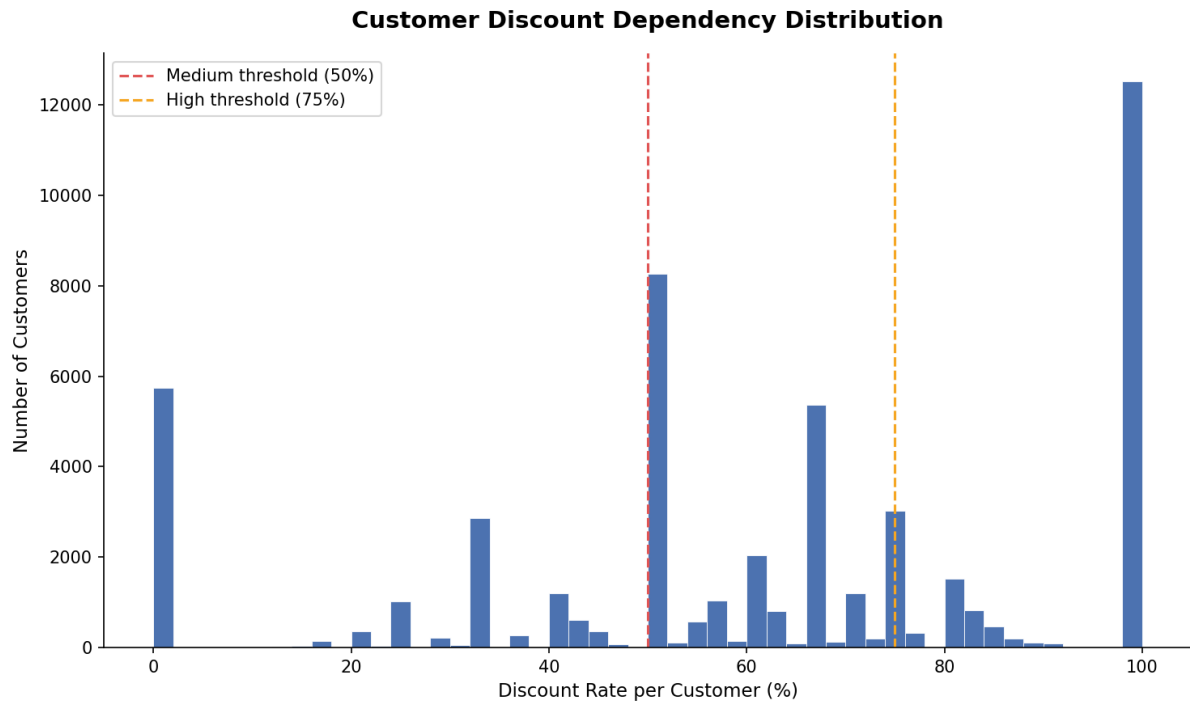


Figure 6. Customer discount rate distribution.

Structural vs. Behavioral Discounting

The distribution of customer discount rates has a mean of 61% and a median of 64%. Approximately 24.2% of customers transact at or above 90% discount usage, while only 11% transact near full price. This polarized distribution rules out selective targeting as the primary driver of high discount prevalence. If discounting were tactical, one would expect a more normal distribution centered around a moderate rate.

Pricing Governance Finding: Discount rates and margin levels are highly consistent across all 74 store locations. This confirms that discounting is centrally governed. Any corrective pricing intervention must be implemented at the policy level to propagate across the full store network.

6. Strategic Recommendations

Recommendation Priority Matrix

PRIORITY	RECOMMENDATION	IMPACT	COMPLEXITY
High	Discount Authorization >10%	High	Medium
High	Weekly Discount Monitoring	High	Low
Medium	Category Discount Ceilings	Medium	Medium
Medium	Controlled Store Pilots	Medium	Medium
Lower	Customer Incentive Migration	Medium	High

Recommendation 1: Introduce Authorization Requirements for Discounts Exceeding 10%

Implementation Horizon

Immediate (0–30d)

Business Problem

Discounting currently affects 61.1% of all transactions and operates as an embedded pricing floor rather than a targeted promotional tool. Because this behavior is consistent across all 74 store locations and product categories, unmanaged discounts routinely reduce operating margins without producing measurable commercial benefit.

Evidence from Analysis

- Discounted transactions generate approximately \$97 less profit per order line compared to full-price transactions.
- Average margin falls consistently from 53.9% in the low-discount tier to 48.9% in the high-discount tier, with no volume recovery at any step.
- The high-discount tier (10-20%) accounts for 79,924 order lines, the largest concentration of discounted volume in the dataset and the primary source of total dollar leakage.
- High-discount customers produce the lowest margins alongside the lowest purchase frequency (3.20 average order lines vs. 6.23 for medium-dependency accounts).

Recommended Action

Establish a centralized authorization requirement for any discount depth greater than 10%, using either a management approval step or an automated flag at the point of sale. Introduce mandatory reason-code tracking so that discount penetration becomes an actively monitored metric rather than a passive output.

Expected Business Impact

Reduced leakage concentrated in the high-volume 10-20% discount tier, improved unit profitability toward the 58.1% full-price baseline, and a shift away from default price concessions in sales negotiations.

Risk / Trade-Off

A strict approval threshold may slow the sales workflow and cause short-term friction for the 24.2% of customers already conditioned to deep discounts [Likely]. Clear turnaround SLAs for authorization approvals would help avoid losing viable transactions.

Recommendation 2: Shift High-Dependency Customers Toward Volume-Based Incentives

Implementation Horizon

Strategic (3–12mo)

Business Problem

High-dependency customers have come to expect discounts on nearly every transaction. This segment returns the lowest individual margins and low purchase frequency, meaning the business is consistently absorbing price reductions without generating proportional commercial return.

Evidence from Analysis

- The high-dependency segment (more than 75% discount usage) covers 16,095 customers transacting at an average individual discount rate of 96.1%.
- This segment yields the lowest average margin at 50.4%, compared to 51.7% for medium-dependency and 53.5% for low-dependency customers.
- High-dependency customers average only 3.20 order lines per account, while medium-dependency customers purchase at nearly twice that rate (6.23 order lines).

Recommended Action

Transition promotional mechanics for high-dependency accounts from flat per-unit discounts to conditional, volume-triggered incentives. Structure discount access around basket-building behaviors, such as minimum line-item counts or cross-sell bundles, to encourage purchasing patterns similar to those observed in the medium-dependency segment.

Expected Business Impact

Reduced margin loss on single-unit orders, increased average order line counts within conditioned customer segments, and a gradual improvement in the profitability of this account group over time.

Risk / Trade-Off

Introducing conditions on discount access may cause an initial drop in transaction volume among customers who resist changing their purchasing behavior [Likely]. This short-term contraction should be modeled against inventory holding targets before rollout.

Recommendation 3: Apply Category-Specific Discount Ceilings to High-Volume Products

Implementation Horizon

Short-term (1–3mo)

Business Problem

Margin compression is consistent across all product categories, but a small number of high-volume categories account for the large majority of total dollar leakage. Applying uniform discount rules across the catalog means the business absorbs the most financial impact precisely where it sells the most, without evidence that the current discount depth is necessary to close those sales.

Evidence from Analysis

- Computers and Cell Phones together drive approximately 46% of all high-discount order lines (18,074 and 17,366 lines respectively).
- Adding the third category (Music / Movies / Audio) brings the combined share to over 65% of all high-discount transactions.
- Margin compression is highly consistent across categories, ranging from 4.9 to 5.3 percentage points, meaning the outsized impact of the top two categories is driven by volume rather than discount depth.

Recommended Action

Set category-specific discount ceilings for Computers and Cell Phones. Rather than removing promotional pricing entirely, test a modest reduction in the maximum allowed discount depth for these categories while keeping existing rules in place for the rest of the catalog. Monitor checkout completion rates during the test to assess price sensitivity.

Expected Business Impact

Margin recovery on the highest-volume product lines, reduced total leakage where the financial impact is most concentrated, and improved pricing discipline without disrupting lower-volume categories.

Risk / Trade-Off

Reducing discounts on popular electronics may slow sales if customers are price-sensitive on these specific items [Likely]. A phased rollout would allow the business to observe demand response before applying changes at full scale.

Recommendation 4: Track Discount Penetration as a Weekly Operational Metric

Implementation Horizon

Immediate (0–30d)

Business Problem

The business currently treats discount penetration as a byproduct of sales activity rather than a managed input. Because the rate has held at approximately 61% year after year without any active adjustment, there is no early warning system to detect margin movement before it appears in monthly financial reports.

Evidence from Analysis

- The overall transaction discount rate has remained stable at 61.1% year-over-year, indicating it functions as an unmanaged baseline rather than a variable the business controls.
- Blended margins are consistent across all 74 stores at approximately 55.9%, confirming that pricing behavior is static and not monitored locally.
- Discounted order lines generate approximately \$97 less profit per line than full-price transactions, a drag that aggregate revenue reporting tends to obscure.

Recommended Action

Configure an automated weekly alert within the existing reporting dashboard to track discount penetration against a defined operating ceiling. A reasonable starting target would be keeping discounted transactions below 50% of weekly order volume. Alerts should notify relevant operators when the rate approaches or exceeds the threshold, prompting a timely review rather than a delayed discovery.

Expected Business Impact

Earlier detection of margin drift, conversion of discount usage into a deliberately managed business input, and better visibility into day-to-day pricing behavior for operational and commercial teams.

Risk / Trade-Off

Setting penetration ceilings too rigidly may put pressure on sales teams during periods of lower demand, where discounting may genuinely support volume [Likely]. These alerts are most effective as conversation starters for review rather than automated enforcement triggers.

Recommendation 5: Test Pricing Policy Changes Through a Controlled Store Pilot

Implementation Horizon

Short-term (1–3mo)

Business Problem

Because discount behavior is centrally governed and nearly identical across all 74 locations, any change to pricing policy carries immediate, network-wide consequences. This makes it difficult to assess how customers will respond to reduced discounting before committing to a full rollout.

Evidence from Analysis

- Discount penetration (approximately 61.1%) and blended margins (approximately 55.9%) are tightly clustered across all 74 stores, confirming that individual locations do not adjust pricing independently.
- No store-level outliers exist, meaning every location currently operates under the same customer discount expectations.
- This uniformity means a small subset of stores can serve as a clean test group, with the remaining stores providing a stable control baseline for comparison.

Recommended Action

Select 3 to 5 store locations to run a 30-day controlled pricing test. Reduce the maximum allowed discount depth by 2 to 3 percentage points within these stores only, while leaving the rest of the network unchanged. Track weekly sales volume and per-line profitability in both groups to measure actual customer response before any broader policy decision is made.

Expected Business Impact

Evidence of real-world price sensitivity without network-wide risk, a reliable basis for evaluating global policy changes, and a clearer understanding of how margin recovery potential translates into actual sales behavior.

Risk / Trade-Off

Customers who regularly shop across multiple locations may notice pricing differences between test and control stores, which could generate confusion or complaints [Likely]. Store staff should be prepared to address pricing questions consistently during the test window.

7. Conclusion

The audit findings are internally consistent and directionally clear. Discounting in this business is not functioning as a targeted promotional mechanism. It is operating as the structural pricing baseline, applied at scale across all categories and all store locations, with no sign of tactical variation or deterioration over time.

The current blended margin of approximately 55.9% reflects the company's actual operating economics under this pricing model. It is not a temporary suppression of a higher baseline. A meaningful portion of the customer base—approximately 24.2% of customers—transact at discount rates exceeding 90%, indicating that discounted pricing has become their expected price point rather than an incentive.

The primary risk is not that margins are declining year-over-year. They are not. The risk is that as transaction volume scales, margin compression scales in direct proportion. Without pricing governance at the policy level, the structural nature of the problem will persist regardless of growth.

8. Appendix: Technical Notes & Assumptions

Metric Definitions

METRIC	DEFINITION
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Discount Flag	Order line where NetPrice != UnitPrice. Validated: no cases exist where NetPrice > UnitPrice.
Discount Depth %	$(\text{UnitPrice} - \text{NetPrice}) / \text{UnitPrice} \times 100$, calculated at line level.
Margin %	$(\text{NetPrice} - \text{UnitCost}) / \text{NetPrice} \times 100$, calculated at line level using quantity-adjusted values.
Profit per Line	Quantity x (NetPrice - UnitCost), summed per order line.
Customer Disc. Rate	Proportion of a customer's order lines where NetPrice != UnitPrice.
Impact Score	Relative index: margin drop (pct pts) x high-discount order line count. Not a monetary value.

Assumptions

- All calculations are performed at the order line level (grain: OrderKey, LineNumber) to avoid aggregation distortion.
- Revenue = Quantity x NetPrice; Cost = Quantity x UnitCost; Profit = Revenue - Cost.
- Currency normalization was not applied. The analysis focuses on relative margin metrics, which reduces sensitivity to currency scale differences.
- Discount tier thresholds (5%, 10%, 20%) are analytically defined for pattern detection and are not derived from internal pricing policy documentation.
- Customer segmentation thresholds (50%, 75%) reflect the approximate P33 and P66 points of the actual customer discount rate distribution.
- Store lifecycle fields (CloseDate, Status) were not used to filter transactions.

Limitations

- Synthetic dataset. Contoso Retail is a benchmark dataset and does not represent a real business. Findings reflect patterns in the data as structured.
- No demand elasticity data. It is not possible to determine whether discounting generates incremental volume or simply reduces price on orders that would have occurred at full price.
- Line-level margin only. Fixed overhead, returns, shipping, and promotional budgets are not captured.
- No loss-making transactions in the dataset. Real-world data may require additional handling.
- Correlation of -0.37 between discount depth and margin confirms a systematic relationship but does not establish that discounting is the sole driver of margin decline.

Tool Stack

TOOL	NOTES
PostgreSQL	v18.3 - SQL audit and profitability analysis
Python	3.13.2 - Pandas, Matplotlib; Jupyter notebooks
Power BI	Desktop - 3-page executive dashboard suite
ReportLab	Python library used to generate this document
Git / GitHub	Version control and repository management